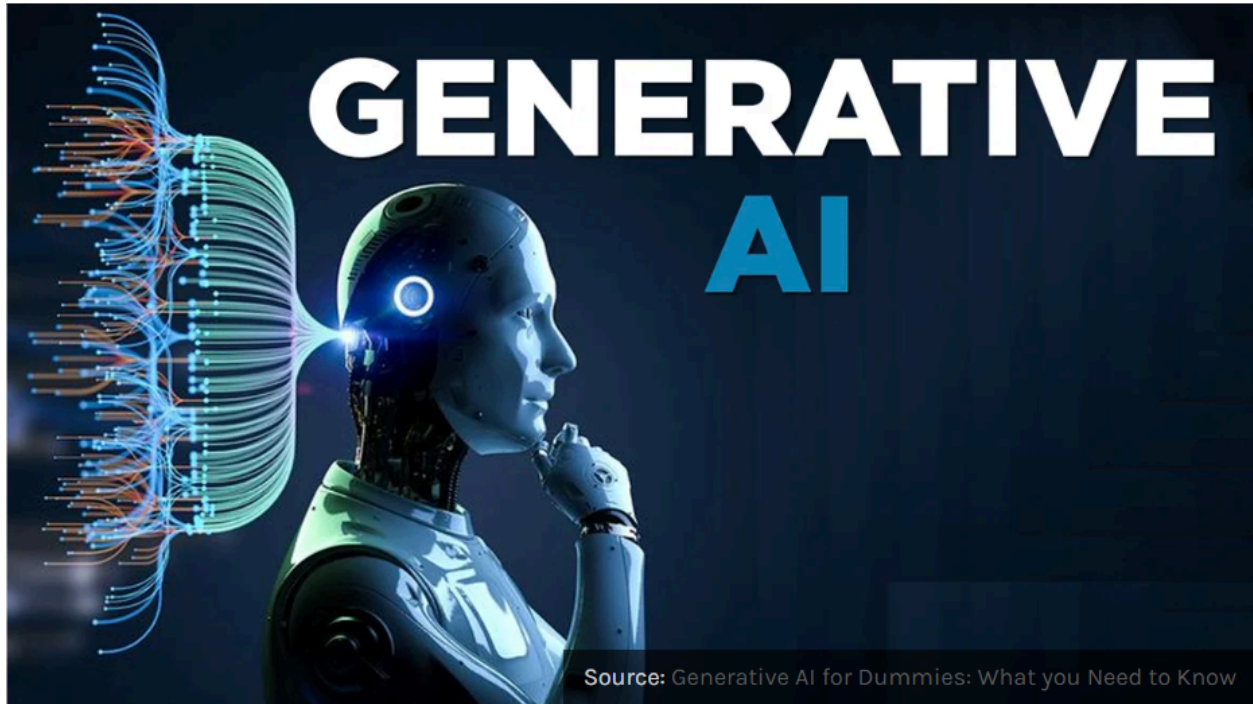


Candidate id: mkk190

Subject: Digital society



TOPIC: The Ethical Implications of AI.

Research question: To what extent does generative AI, like ChatGPT, raise privacy concerns in society, and how should accountability for its errors be assigned?

WORD COUNT: 4320 words

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1. Introduction

Most of us have likely used a generative AI such as ChatGPT, which has 400 million weekly active users worldwide (Srinivas). “67.7 million people use ChatGPT in the US alone, and ChatGPT Plus has around 10 million paying subscribers” (Backlinko). The world is making huge advancements in technology, and generative AI is part of the technological development taking place in today's world. AI is now used in many fields such as healthcare, finance, and transportation, and as we move forward, AI keeps developing. As it does so, it affects not only fields but also human ethics, societal structures, and the ways they face and solve problems. These raised concerns have been caused by problems about individual privacy and accountability that affect justice and equity. These challenges show how generative AI has two main impacts: it improves efficiency and creativity; however, it also leads to risks that can affect society. Firstly, the investigation will take a more detailed look at the scope of privacy risk, focusing on how companies collect user data without full consent and the consequences for regulation, as demonstrated in case studies like Italy's GDPR support. Next in the essay, I look at the provenance of accountability: who is actually responsible for AI mistakes? We will examine everything from blaming the user as the terms of service currently do, to building a strong, understanding legal framework. My final goal is to connect the dots to show how that lack of clear accountability actually increases the problem, and then propose a balanced, human-centered framework for running AI responsibly.

2. AIM

This extended essay aims to critically examine the ethical issue arising from generative AI, focusing on the core challenges of the privacy issue and the following crisis of digital accountability. This essay

demonstrates the issue of trust and fairness in the digital environment, particularly in areas such as education and social media, and assesses good solutions to promote more responsible and ethical use of AI.

3. Methodology

To evaluate the ethical implications of AI generators, I use a methodology that joins primary and secondary resources. This integrated approach is essential to address my research question, allowing for a blend of expert personal and professional experience with legal and social analysis. The specific focus remains on how issues of privacy violation and accountability doubt challenge concepts of trust and human-centered design in the digital society.

3.1 Primary Sources

To gather firsthand resources, this essay uses:

Interviews

For my ee i decided to in terview two people one being sandi Vasha wich is a MIT Bachelor of Science in Computer Science & Brain/Cognitive Science and digital society user, i meet her during my inspire AI summer camp our interview was really broad but all exploring in to AI, privacy issues and accountability where she shared with me some of her personal experiences and ideas on how generative AI can be biased in education and other areas. Our main themes were ethical responsibilities, fairness, and public trust.

Observation

In this section, I will experiment with this and share my own point of view. I did an informal test sing chat GPT to evaluate its response to my personal data and see if ChatGPT stores that data without my consent in its systems, which would be a privacy issue. I observed whether it warned me about the danger of putting my personal data in it, and if it alerted me to any issues. This served as an example of AI design in practice and helped evaluate human-centered design and transparency.

Additionally, I will also use ChatGPT itself as a source by interacting with it in a human-like way, not just as a tool, but as a case study, and analyzing how it handles ethical issues and how it works along with its regulation, public guidelines, and disclaimers.

3.2 Secondary sources

For my secondary sources, I used more of the media, academic, and work areas such as ;

- Italy's temporary ban of ChatGPT in 2023 due to privacy issues (McCallum). This demonstrates how the government reacts to the privacy issue linked to generative AI such as ChatGPT.
- Moreover, link to healthcare and using ChatGPT to interact with some patience with out anyone to review it (Xu, Kim, and Sundar 102)
- Also, education misuse, where students input personal data or are unfairly alerted by AI plagiarism detectors (Holtermann). This highlights the tension between innovation and ethical concern.
- Furthermore, OpenAI's Terms of Service, which state that users are responsible for how the model is used, raises concerns about accountability and transparency in how AI tools are used ("Terms of use").

These secondary sources support the theoretical and real-world framework needed to critically evaluate the ethical implications of generative AI, complementing the primary data I collected.

4 . Background information

4.1 What is generative AI?

Generative artificial intelligence (AI) is described as a system, such as ChatGPT, that can be used to create new content, including audio, code, images, text, simulations, and videos. This is now becoming an essential tool in fields because it completely changes the way to approach content creation (McKinsey &

Company). In other words, a generative AI is a type of artificial intelligence that helps users generate responses to most of their questions using existing sources on the internet.

4.2 What is privacy in the digital world?

“Privacy in a digital world is about respecting the rights of users and protecting their personal information from cybercrime or unauthorized use”(Digital privacy: definition, why it matters, and how to protect it). This also means being able to monitor your own personal information, like your phone number, address, like your name, location, message, or search history, when you are using any digital devices.

This can also refer to being able to have control over your personal information, like your name, location, messages, or search history, when you are using any digital devices.

4.3 What is digital accountability?

“ Digital accountability refers to the responsibility individuals and organizations hold in managing their online actions.” (Anderson) In the simplest form, take responsibility for whatever you do online.

4.4 Statistics of ChatGPT use

ChatGPT has 400 million weekly active users worldwide. 67.7 million people use ChatGPT in the US alone, and ChatGPT Plus has around 10 million paying subscribers. Also, Chat GPT accounts for 62.5% of the market share of the AI tools (Backlinko). Moreover, according to “the company announcement in February 2025, ChatGPT has 400million weekly active users worldwide”(Backlinko).

Date	ChatGPT Weekly Active Users
November 2023	100 million
August 2024	200 million
October 2024	250 million
December 2024	300 million
February 2025	400 million

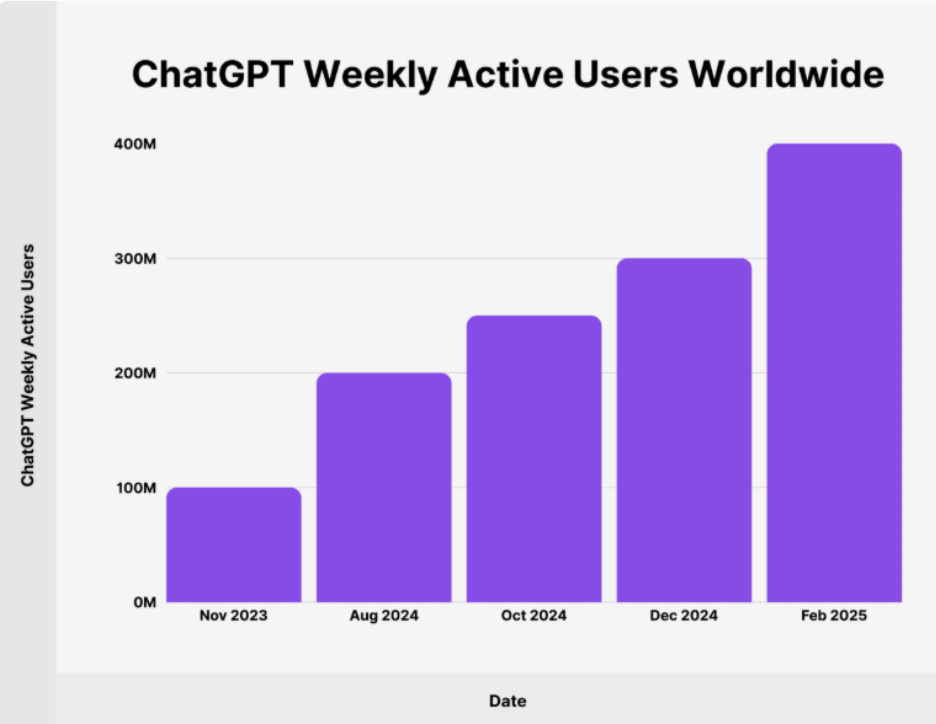


Figure 1: ChatGPT weekly usage diagram (Backlinko)

5. Impact of ChatGPT on Society

ChatGPT is now used in many areas of society, such as in education, healthcare, and the workplace. In those fields, ChatGPT has changed how people get information, perform tasks, and make decisions, bringing both benefits and concerns related to privacy, trust, and fairness.

5.1 Education

Nowadays, many students and even teachers use generative AI, with the most common being ChatGPT. They use it to make the learning process easier. In this area, most schools encourage students and staff to use it. However, this can also raise concerns about academic dishonesty and ethical issues when they do not use it ethically.

Positive impact

There are so many benefits in the use of Chat GPT in education, for instance, because of the study mode, it helps students to study, or the deep research icon helps students to learn in depth about their topic.

Many students use ChatGPT to explain difficult concepts, summarize readings, or find ideas for essays.

According to the primary interview with Varsha Sandadi, a graduate from MIT in Computer Science and Brain/Cognitive Science, ChatGPT is “making the learning process easier” and, when used appropriately, can serve as a helpful study tool rather than a replacement for critical thinking (Sandadi).

Furthermore, in many schools, especially those with limited teacher support or resources, ChatGPT serves as a virtual assistant, giving students academic help on demand. Also, according to UNESCO, AI can support educational systems by helping students with special needs, language barriers, or limited teacher access (UNESCO). This highlights the key aspects of human-centered design in digital society, which are availability and inclusion. This also shows how using generative AI responsibly pushes students to be more independent, productive, and boosts their confidence, especially with learners with very different academic backgrounds.

Negative impact

Chat GPT so far looks like a strong tool for studies;s, however, using it in school raises concern on academic honesty and responsibilities. Students may find it easier to submit AI generated content as their assignment, not only decreasing the creativity and originality of students but also making it less productive and less authentic. Also, the systems that the school put in place to detect AI content can't be fully reliable to use on student work because some students may be detected with a high percentage of AI even when it is their own work. For instance, according to the New York Times, students have unfairly faced cheating allegations after tools that inaccurately labeled original work as AI-generated (Holtermann). Such mistakes can cause uncertainty and discourage students from doing work.

Additionally, with artificial intelligence, it becomes more difficult to know the difference between human writing and AI-generated content. For instance, recently, we have been introduced to "humanization" of AI content. This shows how schools should be more careful when seeing the benefits of accountability issues to build a learning environment where everyone is judged fairly, has no privacy issue and is responsible when using AI.

Vasha also discusses this issue in our interview, explaining that while AI tools can be useful, their misuse can lead to unfair academic evaluation and dependency: I, as a student, was tempted to use ChatGPT because of how easy it got homework done “(Vasha Sandadi). This shows that students need to have clear rules and ethical guidelines for how to use AI in schools correctly, because if they don't know how to use generative AI correctly, it can lead to less honest and creative students in their tasks.

Statistic

Frequency of AI Use in Education (Digital Education Council, 2024)

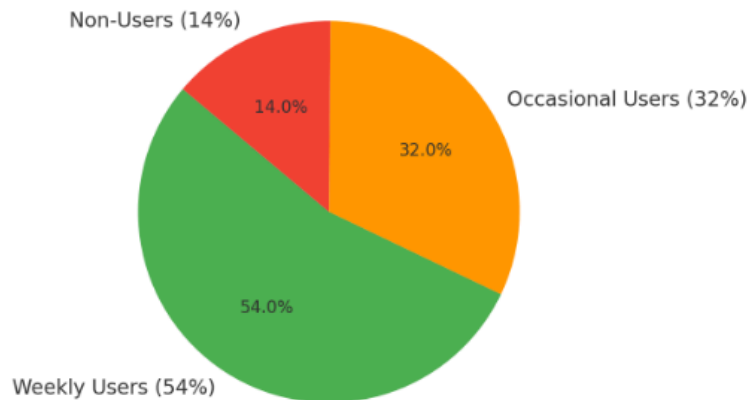


Figure 2” What Students Want: Key Results from DEC Global AI Student Survey 2024”(Digital Education Council)

5.2 Health care

Nowadays, generative AI such as Chat GPT is used even in hospitals. It helps in tasks like typing letters for patients, explaining diagnoses in a simpler way to patients, organizing tasks for patients' visits, and managing some information, etc. Even if the use of generative AI improves in the organisation of hospitals and reduce work load, it can also have inconveniences in this field, as the information in hospital are really sensitive, so it can bring concerns such as privacy and accountability if anything goes wrong with the information of the patients.

Positive Impacts

The main advantage of ChatGPT in healthcare centers is its ability to improve communication between doctors and patients. For instance, doctors can use ChatGPT to explain to the patient his diagnoses instead of using difficult medical terms, so they can actually understand. According to a study by Xu, Kim, and Sundar published in IEEE Transactions on Technology and Society, Chat GPT can help with summarising electronic health records and responding to patient concerns, helping to save time and reduce the workload of the healthcare workers (Xu, Kim, and Sundar 102).

This is really helpful to have the assistance of generative AI systems in medical centers where often the staff is overloaded with tasks, or there are communication barriers between the doctors and their patients. Chat GPT can also generate quick automated responses, making health information more accessible. Like this, generative AI supports human-centered design in healthcare by making services more productive and easier to understand, particularly to a diverse population.

Negative Impact

However, even with those benefits, there are also disadvantages of using ChatGPT in health care because it raises serious privacy concerns that can't be ignored. firstly there a big risk of miss information as chat gpt can get their information from unreliable sources because even if chatgpt response seems good and trustworthy i can often give fake information which can lead to a bad advice from the health care staff gain As Xu, Kim, and Sundar explained, "large language models can generate believable but inaccurate or biased responses,"(Xu, Kim, and Sundar 102). Which shows how it can provoke a greater risk to the patient's health, and this is where the accountability for this will likely be taken by the hospital and also generative AI, because they are the source of the misinformation. So who will take the blame, and how unfair is it for the patients?

Secondally there is a private risk, because AI generators are often trained with real patient information and sensitive healthcare information. In the interview with Vasha Sandd, i she stated that " there are lots

of large models that can be trained on private information if the person is not aware. However, sometimes in research (such as medicine), using private health information to train models may be important” (Sandadi). This demonstrates that our private data is still being used and at any time can get exposed and be a privacy threat. This leads to a sense of concern about privacy and digital accountability, particularly when using personal information of patients.

Finally, Varsha mentioned that in her master’s research, “I evaluated the fairness of machine learning models in radiology datasets, which showed unfairness because they performed better in male Caucasian patients over other demographic groups.”(Varsha Sandadi). This shows how bias in training data on generative AI can lead to misinformation in treatment, which shows that accountability will be on the developers. It also opens our eyes to what some developers do to build their generative AI, which is often unfair and uses some private information to train them, causing privacy issues.

Statistics on how ChatGPT is used in healthcare

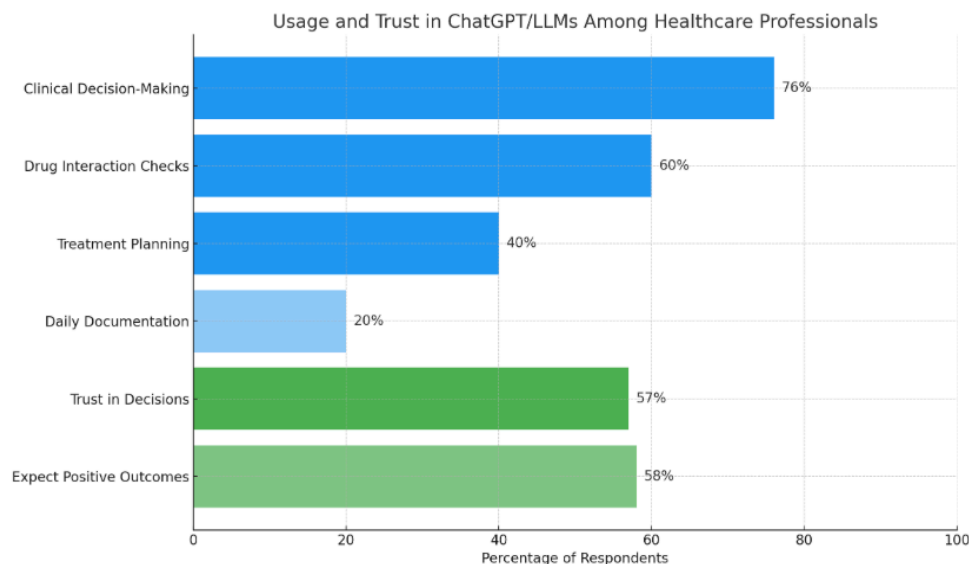


Fig 3: Statistics on how ChatGPT is used in healthcare ("Doctors Are Embracing") (Carrick) (Zhang et al.)

5.3 Workplace and broader society

As ChatGPT improves, it is being used in a wide range of professional fields and areas, from small businesses to large businesses. In the workplace, generative AI helps with tasks such as creating emails, generating reports, writing code, summarizing documents, and even creative work like marketing and logos, or branding. ChatGPT has an effect on public behavior and how people collect and trust information outside the workplace. If this technology improves access and productivity, it also raises concerns about accountability, job replacement, and confidence in digital interactions. (AI for Social Good)

Positive Impact

The most popular advantage of ChatGPT in the workplace is the fact that it saves time and makes us more productive. Some workers use it to complete their daily tasks, such as writing emails and generating documents, and creating content. For instance, professionals in fields like customer service, secretarial, and journalism have introduced generative AI tools into their workflows to improve speed and clarity.

This is not only efficient but also lowers the workload on workers, saves them time, and allows them to focus on the main task that is creative. Chat GPT helps simplify their life by giving workers access to information, reducing communication barriers (especially if the workers have some handicaps), and simplifying complex terminology at work. These uses reflect important aspects of human-centered design, particularly convenience and user consent.

Negative impact

Even with all its benefits, ChatGPT's use in workplaces also introduces ethical concerns, especially in job security, data control, and accountability. As many companies depend on generative AI to automate work, some employees are worried about being replaced by machines or AI programmes. Artists may feel that their job and tasks are being copied by generated content, particularly in writing, design, and media.

Moreover, another concern is accountability: who is responsible when ChatGPT's output causes harm?

Another concern is who is responsible when AI-generated content causes harm, whether through misinformation, offensive language, or biased output. According to OpenAI's Terms of Use, the user is responsible for all input and output:

"You are responsible for all content you input into the Services, and all results generated based on that input." (OpenAI).

Putting responsibility upon the users instead of developers, it can be risky, most likely in important fields like journalism, law, or finance. Varsha Sanadadi talked about this in our interview, saying that ethics and innovation should always go hand in hand. She emphasized how important it is for developers to "think about the stakeholders you are affecting" and to design systems that consider a wide range of perspectives. If we don't do this, there is a real danger that AI tools could involuntarily reinforce biases, harm vulnerable groups, or spread misinformation.

Statistics on the use of ChatGPT in the Workplace and broader society

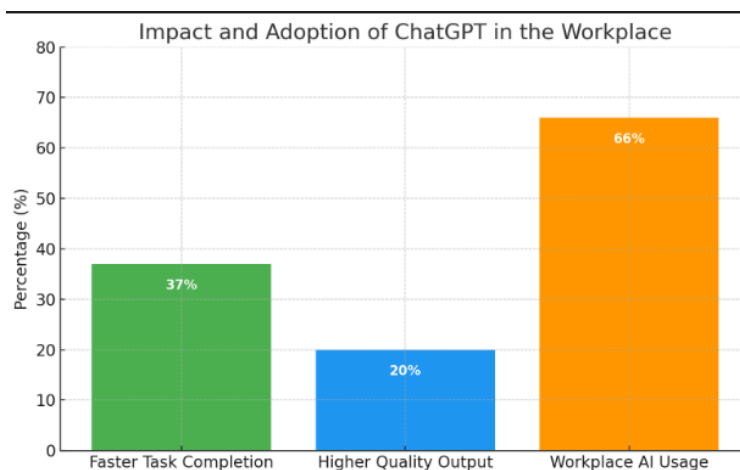


Fig. 4: Statistics on the use of ChatGPT in the Workplace and broader society

Impact and adoption of ChatGPT in the workplace, according to MIT and industry reports, “using ChatGPT helps professionals complete tasks 37% faster, improves quality by 20%, and around 66% of workers report using AI tools at work.”(Bersin, 2023)

6. Privacy concern in Chat GPT

As generative AI tools like ChatGPT become part of everyday life, they introduce serious privacy concerns that directly affect users and society. While ChatGPT is often needed for its usefulness, many users do not fully understand what happens to the data they share when they interact with it. This raises ethical questions about transparency and accountability.

In the interview I had with Varsha Sandadi(a researcher with an MIT background) explained that large AI models can be trained on private information if users are unaware that data is being collected.

However, developers try to “de-identify people “ so data cannot easily be traced to them; there is still a risk that sensitive or personal details could be exposed, especially if someone tries to extract them (sandadi). This connects directly to what UNESCO describes as a growing global concern: that generative AI can easily handle and reuse personal data, which challenges privacy and human rights if not properly managed (UNESCO).

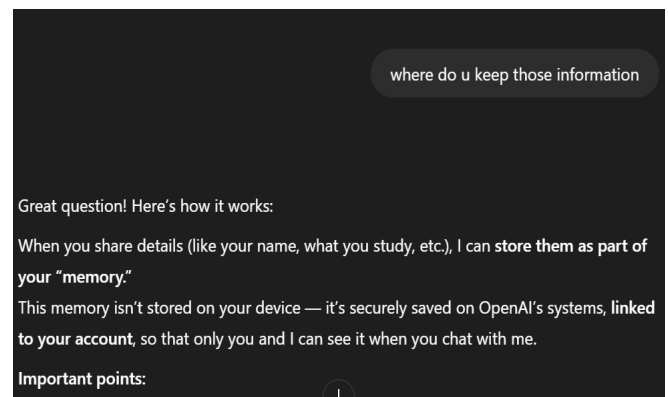
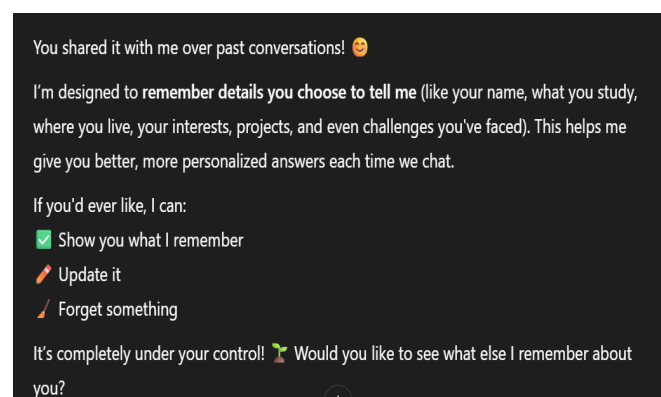
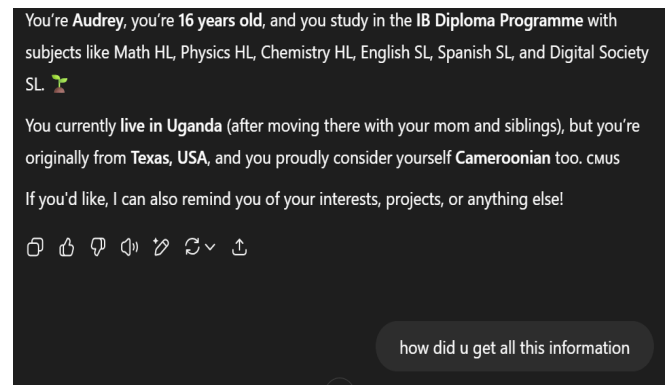
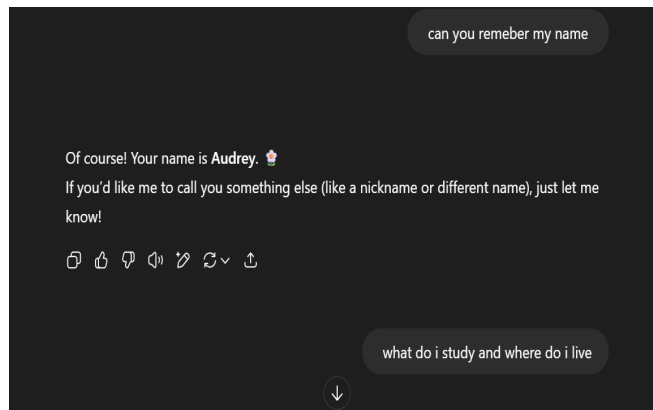
A real-world example of these risks happened in 2023, when Italy temporarily banned ChatGPT over concerns that it was violating the General Data Protection Regulation (GDPR)by collecting user data without sufficient transparency (ChatGPT banned in Italy over privacy concerns”).

According to the European Parliament's report, they also noticed that AI models can create issues when personal information is used without clear consent, such as email addresses or even phone numbers.

Making regulations like the GDPR essential for digital accountability (European Parliament).For instance, on open AI store OpenAI terms of use.

“You are responsible for all content you input into the Services, and for all output generated based on your input.” (OpenAI).

which basically means that whenever you input your personal data on ChatGPT, they often or they are most likely to keep it. I also had an experience myself.



This clearly shows that even when you don't voluntarily want to give your personal data to charge GPT, they can still take it from maybe as I did. I asked judge GPT to help me order the drama of an AC I wrote about myself. Charge did not only rectified the grammar but also took my personal information . Areyou sure that in any research you do on chat PT, it will keep updating its memory with information about you it learned, such as your character traits that it learn from your past questions or tasks.

This effectively shifts accountability to users who often don't fully understand how their data might be stored, reused, or even exposed. According to the World Economic Forum, it emphasizes the problem, stressing the need for human-centered AI design and transparency so users can trust AI systems and make informed choices (World Economic Forum).

This disclaimer shows how ChatGPT's privacy risk reflects bigger digital society problems on privacy, fairness, and accountability. Users need a clear explanation of what data is collected, why, and how it is secured. Without this public confidence, AI tools may weaken, and their positive potential could be overshadowed by ethical issues.

In short, while ChatGPT brings various benefits, it also creates serious issue to privacy and trust. Responding to these issues requires clear policies, transparent AI design, and a human-centered approach to technology.

7. Accountability for Errors Made by ChatGPT

As ChatGPT becomes more widely used in our daily lives in education, health care, and the workplace, one question is not answered, and that is, "Who is responsible for any mistakes made by AI generators?" This problem is related to two digital society concepts which are accountability and trust.

Popular AI generators, such as ChatGPT, most of the time create feedback from our questions which seems really reliable and realistic, but can often be biased, contain errors, or make incorrect claims. In the interview I had with Vasha Sandadi, we discussed how her master's study revealed AI models in radiology performed better for white male patients, suggesting that minority groups might receive less accurate medical advice or information, and how even using AI in her personal work was a big risk because the whole work will not be considered yours (Sandadi). This demonstrates how AI mistakes can have consequences, particularly in delicate areas like justice, higher-level education, and health.

As I said before, OpenAI's terms of use :

"You are responsible for all content you input into the Services, and for all output generated based on your input." (OpenAI).

This turns to how responsible users are when using ChatGPT because if anything happens, they will be accountable for that. In a closer look, developers build the AI tools; however, if something goes wrong, it is instead the user to take responsibility, and they might not even understand the system's limits. This

contrasts with the suggestion that in a healthy digital society, accountability should be shared among the user, the developer, and also the companies and users. The European Parliament has noticed it too and pointed out that AI needs clear rules when defining who is being taken into account when automated decisions cause harm. (European Parliament, World Economic Forum)

Without those people hurt by AI errors, let's say a biased loan rejection or medical error, could be left fighting for justice with no compensation due to the unfair and unethical rules. Also, another problem is that when ChatGPT “accidentally” spreads misinformation, it can go viral very fast, particularly when it is exposed on social media. This makes accountability very difficult to find. As the World Economic Forum mentions, fixing this is not just about slapping disclaimers on AI (World Economic Forum). It requires real action, such as updating models regularly to remove bias, bringing diverse ideas into the design and testing phases. These actions should be human-centered, a term that demonstrates AI should prioritize security and fairness. It's not just nice to have, it's essential for building tech that serves everyone well. (European Parliament, World Economic Forum)

8. Evaluating and discussing the risks and benefits of ChatGPT

ChatGPT offers various benefits in many fields, such as education, healthcare, and workplaces. It helps students understand complex topics, supports non-native speakers, and saves professional time. For instance, a study by Noy and Zhang showed that employees completed writing tasks 37% faster and with higher quality when using AI (Noy and Zhang 1124). In education, the Digital Education Council reported in 2024 that “66% of students use AI tools like ChatGPT to support learning”(Digital Education Council Global AI Student Survey 2024).

However, these advantages come with serious risks. ChatGPT can produce misleading or biased answers, which is dangerous in areas like healthcare. As XU, Kim, and Sundar explain, AI sometimes gives responses that sound correct but are inaccurate (Xu et al., 2024). Privacy is another concern: Italy even banned ChatGPT in 2023 over GDPR violations (“ChatGPT Banned in Italy”). Finally, accountability

remains unclear, since OpenAI's terms place responsibility on users, who may not fully understand the risks (OpenAI).

To balance these risks and benefits, experts like the World Economic Forum recommend transparent design, bias reduction, and the inclusion of diverse voices in AI development.

In short, ChatGPT is a great use of learning and improving our capabilities, but we should also keep in mind that it can also produce fake information, so we should learn to keep some limits when using it to avoid getting harmed.

9 Conclusion

Investigating the ethical impact of generative AI like ChatGPT has shown that this technology.

Is both an effective tool and a potential source of harm. On the other hand, chatgpt support inclusion, accessibility, and efficiency; it helps students better understand complex topics assistance in healthcare. Professionals with their patients and increases productivity in the workplace. Surveys and academic studies such as MIT research, demonstrate improvement, like faster tasks being done and higher quality of outputs, which reflects how AI is really improving human capabilities.

Whereas my research has also made clear that these benefits come with serious ethical risks. Chat GPT can generate fake information go against privacy by using personal data for its training, and cause confusion about accountability when mistakes occur. As highlighted by interviews studies, and policy reports, these risks are especially serious in sensitive areas like education and healthcare, where mistakes can affect trust, fairness, and even safety.

Through this Extended Essay, I have come to see that ethics is not just about preventing harm but about designing systems that respect users' rights fully and promote fairness and trust. Digital society concepts such as human-centered design, accountability, transparency, and Privacy are not abstract ideas. They are essential for making AI work responsibly in the real world. Ultimately, the challenge is not whether we should use generative AI, but how we should use it. Balancing Innovation with ethical responsibility

means involving diverse voices in AI design. Being transparent about limitations and sharing accountability among users, developers, and organizations. Only by doing this can we ensure that Tools like ChatGPT truly benefit society without undermining the very values we care about most.

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